



GAME DEVELOPER

STRATEGY GAME DEMO

HELLO!

At this stage of the interview, you are going to do a small project. The reason behind the project is to show a demo of daily tasks at Panteon to you. Also, it will let us observe and understand your competence and ability on developing games. You can find project details and instructions below.

GENERAL INFORMATION

You are responsible for designing a 2D game with latest 6.3 LTS version of Unity, which includes some basic features of strategy games like building production, unit production and being able to move units towards desired areas. **A playable Windows build should be included in submission.**

1. You have to use all concepts properly which are listed in **"Design"** section while implementing the project.
2. It is expected that there will be various types of buildings and structures besides **"Barracks"**, **"Power Plant"** and **"Soldier Units"**, which are indicated in Production Menu.
3. These buildings can be placed on the map at the suitable locations selected by the user. The user must be informed when there is an inappropriate area. (Eg. such as the red area of the new building.) Also, buildings should have names, images and dimensions.
4. The user should be able to produce as much as he/she wants from the products with no production time (such as barracks, power plants and military units).
5. When the user selects a building in the game board, the image of this building should appear on the information screen. If the selected building is capable of producing units, the images of these products should also be listed.
6. Only barracks can produce units (Soldiers). Power Plants does not produce any products, so no production menu would be needed for Power Plants.
7. Soldiers produced from the barracks must be able to move to the point indicated on the map by using the shortest path (by selecting units with left mouse click then move units with right click) and must wander around the buildings during this journey. There should be a designated spot (aka spawn point) for each barrack where the soldiers will emerge.
8. Barracks will be able to produce 3 types of soldier. All of them should have health points and attack damages.
All of them will have same hp(10) and have different damages per attack.
Soldier 1 -> 10 damage
Soldier 2 -> 5 damage
Soldier 3 -> 2 damage
9. All buildings should have health point.
Barracks -> 100 hp
Powerplant -> 50 hp
10. Selected soldiers will be able to attack with right click on a unit or building.
11. Soldiers and buildings should be destroyed when their hp is 0.
12. Batching and instancing options should be utilized in order to make Draw Call (SetPass Call) count less than 20.
13. Camera placement and UI layout should run smoothly in various aspect ratios and resolutions. Make sure the application window is resizable in Windows build.
14. It is crucial that the code you write is clear and readable. So, you should keep that in mind and stick with the standards. (namings, scalability, comments etc.)
15. Your scene structure, game object namings and folder structure are all important. Please note that your project layout will be reviewed along with your code.
16. The project will be examined in detail. For this reason, you should design your project considering all edge cases.
17. Feel free to use any AI tools to help you write the code. In fact, it's encouraged to have their support. Just remember that we expect you to fully understand every detail in the code.

REFERENCE

We would like to inform you that you are entitled all legal usage rights of each product created by you and made available for trial purposes so that we may recognise your capabilities in accordance with the Law on Intellectual and Artistic Works.

The platform must be 2D.

	PRODUCTION MENU (Infinite Scrollview)	GAME BOARD	INFORMATION
1 Cell = 32 x 32 px. 1 Soldier = 1 x 1 cell Barracks = 4 x 4 cell Power Plant = 2 x 3 cell			

INTERFACE

1. **Production Menu:** An **infinite scroll view** that contains a list of buildings that can be produced.
2. **Game Board:** An area where the produced buildings and soldiers displayed.
3. **Information Panel:** The information area, that belongs to the selected units. (building/soldier)

UX

1. Infinite Scrollview-----Object Pooling

ALGORITHM

1. Pathfinding (A*)

SAVE & SEND!

You need to share your complete project via GitHub (send us an email that contains the link to the project – hr@panteon.games).

GOOD LUCK AND HAVE FUN!

LEGAL NOTICE

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